Question:
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?
Answer Choices:
A. 2.48 × 10^-8 cm
B. 5.92 × 10^7 cm/sec
C. 8.66 × 10^-6 cm
D. 3.14 × 10^-5 cm
E. 9.81 × 10^-4 cm
F. 5.27 × 10^-9 m
G. 1.23 × 10^-7 cm
H. 4.14 × 10^-10 m
I. 6.625 × 10^-27 erg sec
J. 0.512 × 10^6eV

Reasoning + Answer:

<reasoning process here>
We are asked to find the de Broglie wavelength of an electron with a kinetic energy of 1 eV.

The de Broglie wavelength is given by:

Î» = h / p

where:
- h = Planck's constant = 6.626 ÃĹ10âģ»Â³âģ´ JÂ·s
- p = momentum of the electron

For a non-relativistic electron, kinetic energy K = (1/2) mvÂ², and momentum p = mv.

We can relate kinetic energy and momentum:

p = âĪļ(2mK)

So,

Î» = h / âĪļ(2mK)

We need to compute this for K = 1 eV.

First, convert 1 eV to joules:

1 eV = 1.602 ÃĹ10âģ»Â¹âģ¹ J

Mass of electron m = 9.109 ÃĹ10âģ»Â³Â¹ kg

Now compute:

Î» = h / âĪļ(2mK)

Plug in the values:

Î» = (6.626 ÃĹ10âģ»Â³âģ´) / âĪļ(2 ÃĹ9.109 ÃĹ10âģ»Â³Â¹ ÃĹ1.602 ÃĹ10âģ»Â¹âģ¹)

First compute the product inside the square root:

2 ÃĹ9.109e-31 ÃĹ1.602e-19 = 2 ÃĹ9.109 ÃĹ1.602 ÃĹ10âģ»âģµâģ°
âīĪ2 ÃĹ14.6 ÃĹ10âģ»âģµâģ° = 29.2 ÃĹ10âģ»âģµâģ° = 2.92 ÃĹ10âģ»âģ´âģ¹

Now take square root:

âĪļ(2.92 ÃĹ10âģ»âģ´âģ¹) âīĪ âĪļ2.92 ÃĹ10âģ»Â²âģ´.âģµ âīĪ1.71 ÃĹ10âģ»Â²âģ´.âģµ

Better to compute numerically:

âĪļ(2.92e-49) = âĪļ(2.92) ÃĹ10âģ»Â²âģ´.âģµ âīĪ1.71 ÃĹ10âģ»Â²âģ´.âģµ

But 10âģ»Â²âģ´.âģµ = 10âģ»Â²âģ´ ÃĹ10âģ»âģ°.âģµ âīĪ10âģ»Â²âģ´ ÃĹ0.316 = 3.16 ÃĹ10âģ»Â²âģµ

Wait âĢĶ better:

âĪļ(2.92 ÃĹ10âģ»âģ´âģ¹) = âĪļ2.92 ÃĹ âĪļ10âģ»âģ´âģ¹ = 1.71 ÃĹ10âģ»Â²âģ´.âģµ

But 10âģ»Â²âģ´.âģµ = 10âģ»Â²âģ´ ÃĹ10âģ»âģ°.âģµ âīĪ10âģ»Â²âģ´ ÃĹ0.3162 = 3.162 ÃĹ10âģ»Â²âģµ

So âĪļ(2.92 ÃĹ10âģ»âģ´âģ¹) âīĪ1.71 ÃĹ3.162 ÃĹ10âģ»Â²âģµ? No.

Wait: âĪļ(a ÃĹ10âģ»âģ´âģ¹) = âĪļa ÃĹ10âģ»Â²âģ´.âģµ

âĪļ(2.92 ÃĹ10âģ»âģ´âģ¹) = âĪļ2.92 ÃĹ10âģ»Â²âģ´.âģµ

âĪļ2.92 âīĪ1.71

10âģ»Â²âģ´.âģµ = 10âģ»Â²âģ´ ÃĹ10âģ»âģ°.âģµ